Day - 12

Quantum Teleportation Protocol & Circuit Design

Welcome to Day 12 of your 30-Day Quantum Computing Challenge! Today, we explore **Quantum Teleportation**—a protocol that combines superposition, multi-qubit systems, and entanglement to achieve an extraordinary communication task.

1. Why We Need Quantum Teleportation

Imagine you have prepared an unknown qubit in a highly specific state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and you want to transmit this exact state to a processor hundreds of kilometers away. Your classical intuition might say: *"Just copy it and send the data."*

However, quantum physics forbids this due to the **No-Cloning Theorem**.

The No-Cloning Theorem

An unknown, arbitrary quantum state cannot be copied or duplicated perfectly. If you attempt to measure the state to read its amplitudes (α and β), you trigger a **state collapse**, destroying the original quantum information forever.

Quantum Teleportation solves this fundamental constraint. Instead of physically transporting the original particle or making a duplicate copy, the protocol **transfers the exact quantum state** from a source qubit to a distant target qubit. No copies are ever created; the original state is systematically disassembled (destroyed) at the sender's end and perfectly reconstructed at the receiver's end.

2. Core Mechanism: The Three Ingredients

To execute the protocol, Alice (the sender) and Bob (the receiver) must manage three physical qubits and three essential resources:

The Three Qubits

1. **Qubit 0 (q_0):** The unknown state $|\psi\rangle$ that Alice wants to send. Neither Alice nor Bob knows its exact parameters.
2. **Qubit 1 (q_1):** Alice's half of a pre-shared entangled Bell pair.
3. **Qubit 2 (q_2):** Bob's half of the pre-shared entangled Bell pair.

The Three Essential Resources

- **The Source Qubit:** Containing the active state to be transferred.

- **A Shared Entangled Pair:** Acting as the *quantum communication channel*.
- **Two Classical Bits (c):** Used to transmit Alice's measurement results to Bob.

3. The Teleportation Circuit Architecture

The protocol is visually structured as a parallel workflow moving left to right across time:

```
                 [ State Prep ]  [ Alice's Operations ]   [ Readout ]
|psi> (q0) ---------*------------------| H |---------| M
   |-------||------
                    |                                |___|       ||
                [ Bell Prep ]                                    ||
                   c0
|0> (q1) ---| H |---X---|-----------------------------| M
   |-------||---
            |___|       |                            |___|       ||
                 ||
                        |                                        ||
                              || c1
|0> (q2)
   ---------------*-----------------------------------------||---||--

                                                                 ||


                                                                    ||

                                                            .---++---++---.

                                                            | Bob'
                                                               s

                                                               |
                                                            |
                                                               Correction
                                                                   |
                                                            '---|---------'

                                                                |
Bob's Recovered State |psi>
   <------------------------------------+------
```

Protocol Workflow Step-by-Step

Step 1: Pre-Shared Entanglement
Alice and Bob generate a maximally entangled Bell state $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ between q_1 and q_2. Alice holds q_1, and Bob takes q_2 to his remote location.

Step 2: Alice's Entangling Operations
Alice applies a CNOT gate with her unknown qubit (q_0) acting as control and her Bell qubit (q_1) as the target. She then passes q_0 through a Hadamard (H) gate. This spreads the hidden quantum information across the entire multi-qubit system.

Step 3: Measurement & State Destruction
Alice performs a standard Z-basis measurement on her two qubits (q_0 and q_1). This

action triggers state collapse and converts the quantum parameters into **two classical bits** (c_0 and c_1). Crucially, this measurement **destroys** the original state at Alice's end, adhering to the No-Cloning Theorem.

Step 4: Classical Communication (The Relativity Shield)

Alice transmits the 2 classical bits to Bob over a standard classical channel (fiber optics, radio waves, etc.). Because Bob cannot decipher or reconstruct his qubit without these 2 classical bits, information **cannot** travel faster than the speed of light.

Step 5: Bob's Conditional Corrections

Once Bob receives the classical bits, he looks up the required correction gate to re-align his qubit's vector:

Alice's Bits ($c_1 c_0$)	**Bob's State**	**Correction Gate**	**Reconstructed State**
00	$\alpha\|0\rangle + \beta\|1\rangle$	**I (Do Nothing)**	$\alpha\|0\rangle + \beta\|1\rangle$
01	$\beta\|0\rangle + \alpha\|1\rangle$	**X Gate (Bit-Flip)**	$\alpha\|0\rangle + \beta\|1\rangle$
10	$\alpha\|0\rangle - \beta\|1\rangle$	**Z Gate (Phase-Flip)**	$\alpha\|0\rangle + \beta\|1\rangle$
11	$\beta\|0\rangle - \alpha\|1\rangle$	**X then Z Gate**	$\alpha\|0\rangle + \beta\|1\rangle$

Once Bob executes the correction, his physical qubit holds the exact state vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ that Alice started with.

4. Complete Program Implementation in Qiskit

This script builds a 3-qubit register, prepares an initial test state ($|+\rangle$), conducts the teleportation sequence, and applies conditional logic (`c_if`) based on classical registers.

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

# Initialize circuit architecture: 3 qubits, 2 classical bits
qc = QuantumCircuit(3, 2)

# --- STEP 1: Prepare the state to teleport ---
# Sending a |+> state for demonstration purposes
qc.h(0)

# --- STEP 2: Create the shared Bell pair ---
qc.h(1)
qc.cx(1, 2)

# --- STEP 3: Alice's entangling operations ---
qc.cx(0, 1)
qc.h(0)

# --- STEP 4: Measurement into classical registers ---
qc.measure(0, 0)
qc.measure(1, 1)

# --- STEP 5: Bob's conditional corrections ---
# Bob applies X if c_0 is 1, and Z if c_1 is 1
qc.x(2).c_if(qc.cregs[0], 1)
qc.z(2).c_if(qc.cregs[0], 2)

# Initialize the simulator backend
sim = AerSimulator()
compiled_circuit = transpile(qc, sim)

# Execute the circuit workload
result = sim.run(compiled_circuit, shots=1024).result()
counts = result.get_counts()
print("Execution␣Measurement␣Counts:", counts)
```

5. Summary Comparison Matrix

Core Metric	Classical Information Transfer	Quantum Teleportation Protocol
Primitive Unit	Duplicates bits sequentially across channels.	Transfers state vector tracking variables.
State Duplication	Unlimited identical copies can coexist.	Original state **must** be completely destroyed.
Channel Requirements	Single classical data line.	Pre-shared entanglement + classical link.
Transmission Speed	Capped at the speed of light.	Instant correlation changes, but completion is capped at light speed.
Hardware Use Case	Text, media, legacy data networks.	Quantum Networks, repeaters, and distributed systems.